HBOT Use Cases, Protocols, and Scientific References

Overview

This document compiles hyperbaric oxygen therapy (HBOT) use cases, protocol patterns, and scientific references for long COVID, fibromyalgia, dermatology, plastic surgery, and women's health. The strongest evidence in this set is for long COVID symptom improvement, fibromyalgia, compromised grafts/flaps, and difficult wound healing, while women's health uses such as fertility and endometriosis remain investigational.[1][2][3][4][5]

Long COVID

Randomized and controlled studies suggest HBOT may improve fatigue, cognition, sleep, pain, and quality of life in selected long COVID patients. One trial protocol used up to 10 sessions over 6 weeks, while later controlled work commonly used 40 sessions at around 2.0 ATA with 100% oxygen for 60 to 90 minutes.[6][7][8][9][1]

Protocol patterns

- 2.0 ATA, 60 to 90 minutes, typically 10 to 40 sessions.[9][6]

- Often delivered 5 days per week in intensive protocols.[7][6]

- Primary endpoints include fatigue, brain fog, sleep, pain, and quality-of-life measures.[6][7]

Reference links

- [Does Hyperbaric Oxygen Therapy (HBOT) Help Treat Long COVID?](https://www.yalemedicine.org/news/does-hyperbaric-oxygen-therapy-hbot-help-treat-long-covid)[6]

- [Efficacy and safety of hyperbaric oxygen therapy for long COVID](https://pmc.ncbi.nlm.nih.gov/articles/PMC11138265/)[1]

- [Hyperbaric oxygen for treatment of long COVID-19 syndrome (HOT-LoCO): trial protocol](https://bmjopen.bmj.com/content/12/11/e061870)[9]

- [Long term outcomes of hyperbaric oxygen therapy in post covid condition](https://www.nature.com/articles/s41598-024-53091-3)[7]

- [Hyperbaric oxygen effectively addresses the pathophysiology of long COVID: clinical review](https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2024.1354088/full)[8]

Fibromyalgia

The most cited non-TBI fibromyalgia study used 40 sessions at 2 ATA for 90 minutes over 8 weeks and reported significant improvement in pain, tender points, quality of life, and brain activity patterns. Meta-analyses published in 2023 found improvements in function, tender points, fatigue, sleep, and quality of life, although the magnitude of pain improvement varied between pooled analyses.[10][11][12][13][14]

Protocol patterns

- 2.0 ATA, 90 minutes, 40 sessions, usually 5 days per week for primary fibromyalgia.[11][10]

- Some protocols use 60 sessions for more refractory populations, though these often overlap with TBI-associated fibromyalgia literature.[15]

- Lower-pressure approaches have also been explored, but the best-known primary fibromyalgia protocol remains 2 ATA for 40 sessions.[13][16]

Reference links

- [Hyperbaric Oxygen Therapy Can Diminish Fibromyalgia Syndrome](https://pmc.ncbi.nlm.nih.gov/articles/PMC4444341/)[10]

- [Hyperbaric oxygen therapy compared to pharmacological intervention in fibromyalgia associated with TBI](https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0282406)[15]

- [Effectiveness of Hyperbaric Oxygen for Fibromyalgia: A Meta-analysis](https://pmc.ncbi.nlm.nih.gov/articles/PMC10204569/)[12]

- [Efficacy and safety of hyperbaric oxygen therapy for fibromyalgia](https://pmc.ncbi.nlm.nih.gov/articles/PMC9872467/)[13]

- [Systematic review and meta-analysis in BMJ Open](https://bmjopen.bmj.com/content/13/1/e062322)[14]

Dermatology

In dermatology, HBOT is most strongly supported for chronic wounds, radiation injury, burns, compromised grafts/flaps, and necrotizing infection rather than for cosmetic rejuvenation. Evidence for inflammatory dermatoses such as atopic dermatitis and psoriasis is limited to small studies, case series, and case reports.[17][18][19][20][21][22]

Main use cases

- Chronic and nonhealing wounds, including diabetic and vascular ulcers.[18][17]

- Radiation dermatitis and delayed soft-tissue healing after radiotherapy.[23][17]

- Compromised skin grafts and flaps.[19][17]

- Burns and necrotizing soft-tissue infection.[24][17]

- Early-stage applications in atopic dermatitis, psoriasis, hidradenitis suppurativa, livedoid vasculopathy, and pyoderma gangrenosum.[20][17][23]

Protocol patterns

- 2.0 to 2.5 ATA, 60 to 90 minutes, commonly 20 to 40 sessions for chronic wounds and radiation injury.[19][23]

- More urgent daily treatment for compromised grafts/flaps when perfusion issues are identified.[25][19]

- In inflammatory skin disease, reported protocols vary widely and are not standardized.[20][23]

Psoriasis and eczema studies

Psoriasis evidence is based mainly on case reports. One 2009 report described complete remission of pustular and arthropathic psoriasis after 6 sessions at 2.8 ATA for 60 minutes, while another psoriasis vulgaris case improved after 15 sessions at 2.0 ATA for 90 minutes. In atopic dermatitis, a pediatric study reported improvement in severe disease after a 30-session cycle, and a 2025 review described reductions in pruritus, lesion severity, and Staphylococcus aureus colonization.[21][22][26][20]

Reference links

- [The use of hyperbaric oxygen therapy in the treatment of skin diseases](https://acta-apa.mf.uni-lj.si/journals/acta-dermatovenerol-apa/papers/10.15570/actaapa.2019.20/actaapa.2019.20.pdf)[17]

- [Role of Hyperbaric Oxygen Therapy in Cosmetic and Therapeutic Skin Conditions](https://pmc.ncbi.nlm.nih.gov/articles/PMC9748824/)[18]

- [Hyperbaric Therapy for Skin Grafts and Flaps](https://www.ncbi.nlm.nih.gov/books/NBK470219/)[19]

- [The Supporting Role of Hyperbaric Oxygen Therapy in Atopic Dermatitis](https://pmc.ncbi.nlm.nih.gov/articles/PMC12072933/)[20]

- [Therapeutic effect of hyperbaric oxygen in psoriasis vulgaris](https://pmc.ncbi.nlm.nih.gov/articles/PMC2737769/)[21]

- [Effects of Hyperbaric Oxygen Therapy in Children with Severe Atopic Dermatitis](https://pmc.ncbi.nlm.nih.gov/articles/PMC8001365/)[22]

Plastic surgery and wound healing

HBOT is best supported in plastic and reconstructive surgery as an adjunct for compromised grafts/flaps, difficult postoperative wounds, radiation-injured tissue, burns, and traumatic soft-tissue injury. It is most defensible for tissue salvage and impaired healing, rather than as a routine enhancer for uncomplicated cosmetic surgery.[27][28][19]

Protocol patterns

- 2.0 to 2.5 ATA, 60 to 90 minutes, usually once daily.[28][19]

- Early initiation when ischemia or venous congestion is suspected in a flap or graft.[25][19]

- Shorter postoperative support courses, often around 5 to 10 sessions, are reported in selected aesthetic cases.[29][30]

Evidence highlights

A facelift study reported mean healing of 13.3 days in the HBOT group versus 36.9 days in controls, with the first session started within 24 hours after surgery and sessions averaging about 78 minutes at around 2.0 ATA. Reviews of compromised flaps and grafts describe HBOT as a salvage adjunct that improves oxygen delivery, angiogenesis, fibroblast activity, and edema control while standard surgical management addresses the mechanical cause of failure.[3][4][19]

Reference links

- [Hyperbaric oxygen therapy in Plastic Surgery practice](https://pubmed.ncbi.nlm.nih.gov/32011977/)[27]

- [Hyperbaric Therapy for Skin Grafts and Flaps](https://www.ncbi.nlm.nih.gov/books/NBK470219/)[19]

- [Hyperbaric Oxygen Therapy for the Compromised Graft or Flap](https://pmc.ncbi.nlm.nih.gov/articles/PMC5220535/)[4]

- [Assessing the Efficacy of Hyperbaric Oxygen Therapy on Facelift Recovery](https://pmc.ncbi.nlm.nih.gov/articles/PMC10387739/)[3]

- [Enhancing Postoperative Healing with Hyperbaric Oxygen Therapy](https://www.intechopen.com/chapters/1197117)[29]

- [Applications of Hyperbaric Oxygen Therapy in Plastic Surgery](https://medical-clinical-reviews.imedpub.com/articles/applications-of-hyperbaric-oxygen-therapy-in-plastic-surgery.php?aid=48456)[28]

Women’s health

The strongest women’s-health signal is in fertility-related applications and endometriosis, but these uses remain investigational and are not established standards of care. No strong clinical evidence was identified for menopause symptoms such as hot flashes, sleep disturbance, or mood symptoms as a validated HBOT indication.[5][31][32][27]

Fertility

Published fertility-related literature suggests HBOT may influence ovarian function, oocyte quality, and embryo outcomes in selected settings, including IVF-related treatment windows. Reported protocols include around 2.3 ATA for about 70 minutes in cycle-based treatment schedules, but there is no universally accepted fertility protocol.[31][33][5]

Endometriosis

Endometriosis is an emerging application with early clinical and translational support. A pilot study described symptom improvement after 20 sessions over 4 weeks, and a registered trial indicates the topic is under active investigation.[32][34][35]

Menopause symptoms

The available literature found here does not support menopause symptom treatment as a well-established HBOT indication.[5][27]

Reference links

- [Investigational HBOT Indications - Infertility](https://woundreference.com/p/blog?id=investigational-hbot-indications-infertility)[5]

- [Hyperbaric Oxygen Treatment Ameliorates the Decline in Oocyte Quality](https://pmc.ncbi.nlm.nih.gov/articles/PMC9753892/)[31]

- [Hyperbaric oxygen therapy improves oocyte yield and embryo outcomes](https://pmc.ncbi.nlm.nih.gov/articles/PMC12621376/)[33]

- [HBOT for Endometriosis](https://oxify.co.uk/health-condition/hbot-for-endometriosis/)[34]

- [Breathing New Life Into Endometriosis Treatment](https://www.gynecologiconcologyinstitute.org/news/endometriosis/endometriosis-research/hyperbaric-oxygen-treatment-endometriosis)[35]

- [The HEROES Trial: Hyperbaric Oxygen Therapy for Endometriosis](https://clinicaltrials.gov/study/NCT06579040)[32]

Practical positioning

For clinical communication, the most evidence-based HBOT positioning across these topics is: long COVID symptom support, fibromyalgia, compromised flaps/grafts, difficult wound healing, radiation injury, and selected dermatologic wound applications. Fertility and endometriosis should be labeled investigational adjuncts, while menopause symptom claims should be avoided unless stronger clinical evidence emerges.[1][10][17][27][32][5][19]

Sources

[1] Efficacy and safety of hyperbaric oxygen therapy for long COVID
https://pmc.ncbi.nlm.nih.gov/articles/PMC11138265/

[2] Hyperbaric Oxygen Therapy Can Diminish Fibromyalgia Syndrome – Prospective Clinical Trial
https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0127012

[3] Assessing the Efficacy of Hyperbaric Oxygen Therapy on Facelift ...
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[4] Hyperbaric Oxygen Therapy for the Compromised Graft or Flap - PMC
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[8] Frontiers | Hyperbaric oxygen effectively addresses the pathophysiology of long COVID: clinical review https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2024.1354088/full

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[16] Low-pressure Hyperbaric Oxygen Therapy and Physical Exercise ... https://oxynova.com/2021/09/13/low-pressure-hyperbaric-oxygen-therapy-and-physical-exercise-protocol-in-women-with-fibromyalgia/

[17] The use of hyperbaric oxygen therapy in the treatment of ... https://acta-apa.mf.uni-lj.si/journals/acta-dermatovenerol-apa/papers/10.15570/actaapa.2019.20/actaapa.2019.20.pdf

[18] Role of Hyperbaric Oxygen Therapy in Cosmetic and ... - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC9748824/

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[23] Dermatology - HBOT Clinic Galway Ireland https://www.oxygeneration.com/hbot-resource-library/dermatology

[24] 05 Piotrowska ANG.indd https://publisherspanel.com/api/files/view/1735629.pdf

[25] Utilizing Hyperbaric Oxygen Therapy for Optimal Healing of ... https://www.o2oasis.com/utilizing-hyperbaric-oxygen-therapy-for-optimal-healing-of-compromised-skin-grafts-and-flaps/

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